BLOCKCHAIN INDIA CHALLENGE · 2026

MINISTRY OF ELECTRONICS & INFORMATION TECHNOLOGY · C-DAC

MerkNet

A permissioned blockchain for cross-border manufacturing trade

SUBMISSION AT A GLANCE

Category	Stage	Submitting entity	Proposition
Supply Chain Management	Prototype Stage Review	Merkantis Private Limited	Digitising India - Europe trade documentation on Hyperledger Fabric, signed with existing DSCs.

SUBMITTED BY

Divyansh Malhotra

Founder, Merkantis Pvt Ltd

divyansh@merkantis.com

NON-CRYPTO DECLARATION · *MerkNet uses zero cryptocurrency, zero mining, zero tokens. Built on Hyperledger Fabric 3.0+ (Apache 2.0), FireFly (Apache 2.0), and IPFS (MIT). Full compliance in Section 9.*

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1. Executive Summary

Cross-border manufacturing sourcing from India fails not because of bad suppliers, but because no intermediary owns the outcome and the documentation infrastructure is fragmented, manual, and trust-deficient. This is not a technology problem in isolation - it is a trust infrastructure problem that blockchain is uniquely positioned to solve.

Indian MSMEs generate roughly **$197 billion** in merchandise exports every year, yet every cross-border shipment still moves on paper. The cost is measurable: **3–5 days per order lost** to document preparation, a **15–20% discrepancy rate** at destination ports, and demurrage charges of **$500 - 2,000 per incident**. For MSMEs on 8–12% margins, this overhead is the difference between profitability and loss.

The infrastructure to fix this already exists. India's CCA-accredited Digital Signature Certificates are X.509 certificates - the exact identity standard Hyperledger Fabric uses. Foreign Trade Policy 2025 explicitly mandates blockchain for trade records. The Bills of Lading Act 2025 grants full legal recognition to electronic Bills of Lading. The statutory, cryptographic, and technical foundations are in place. What is missing is a production platform that connects them.

MerkNet is that platform. A permissioned, non-crypto blockchain on which every party in a cross-border order - manufacturer, inspector, freight forwarder, customs broker, bank, and buyer - shares a single tamper-proof record. Documents are signed with existing DSCs. Any party can verify any document in under five seconds. No phone calls, no email chains, no disputed signatures.

> *"Merkantis is an export execution company, not a blockchain vendor. The Prototype runs on orders already in our pipeline - no cold-start, no synthetic data, no hypothetical users."*

Dimension	Today	MerkNet Target	Measurement
Document preparation time	3–5 days	Under 4 hours	Timestamp: request to verified on-chain
Discrepancy rate at destination	15–20%	0% on pilot orders	Buyer confirmation at port
Signature verification	1–3 days	Under 5 seconds	On-chain hash lookup latency
Supplier onboarding	2–4 weeks	Under 8 hours	Invitation to first signed transaction

KEY TAKEAWAY **MerkNet is the production implementation of statutes India has already enacted, built by the team that feels the operational pain every day.**

2. Problem Statement

A typical cross-border manufacturing order passes through six to eight parties: manufacturer, quality inspector, freight forwarder, customs broker, insurance provider, bank, regulator, and buyer. Each party maintains a private copy of every document. There is no shared record, no verifiable chain of custody, and no common source of truth. The downstream costs compound at every handoff.

Failure mode	Operational impact
Manual signature verification	Agencies and banks phone issuing authorities to validate signatures. 3–5 days per order, every order.
Document discrepancies at destination	15–20% of shipments face mismatches. Demurrage averages $500–2,000 per incident. Repeat-order conversion erodes.
Unverifiable chain of custody	Quality certificates are PDFs in email. Signatures can be visually spoofed. Liability is unresolvable after the fact.
Duplicate financing fraud	The same Bill of Lading can be presented to multiple banks. No shared ownership record prevents double-financing.
Verbal commitments	Delivery timelines and specifications exist only in WhatsApp threads. No immutable record of what was promised versus delivered.

Why this matters for India. MSME exporters compete directly with Chinese and Vietnamese suppliers who operate on thinner documentation overhead. A Coimbatore precision parts manufacturer shipping CNC components to a Swiss buyer loses five to ten working days per order to paperwork. That lost time is lost margin, lost orders, and lost competitive position. Fixing it is not a productivity improvement - it is a structural lever for export competitiveness.

There is a second, less-discussed failure mode. Indian suppliers routinely quote optimistic timelines and delay problem notification. This is culturally rational but operationally destructive for buyers on fixed production schedules. Without a shared, immutable record of committed milestones versus actual delivery, disputes resolve on the basis of whoever argues loudest. Blockchain does not fix the culture, but it does fix the accountability gap.

KEY TAKEAWAY **The problem is structural, measurable, and directly erodes India's export competitiveness. It is also ready to be solved with infrastructure that already exists.**

3. Solution Architecture

MerkNet is a permissioned Hyperledger Fabric network on which the full lifecycle of a cross-border order - from RFQ through payment reconciliation - is recorded as a sequence of cryptographically signed, tamper-proof transactions. Documents are anchored by hash on-chain and stored encrypted off-chain on IPFS. Identity is derived from India's existing DSC infrastructure. The system is operational from day one because every building block already exists in production form, we only need to connect them in our custom configuration.

3.1 Core operating principles

- **One record, all parties.** Six to eight private copies collapse into one shared ledger entry visible only to authorized participants via Fabric channels and private data collections.
- **Signature verification in seconds.** DGFT officers, customs officials, and accredited inspectors sign with their existing Class 3 DSCs. Any peer validates cryptographically - no callback, no latency.
- **Zero-infrastructure onboarding.** MSMEs do not run servers or manage keys. Merkantis operates managed identity with time-bounded custody; participants graduate to self-managed keys after their first completed deal.
- **QR verification everywhere.** Every on-chain document carries a QR code verifiable on a phone in under five seconds.
- **Pre-submission compliance.** The Compliance chaincode validates documents against DGFT policy and destination-country regulations (CE, REACH, RoHS) before submission. Errors are caught at source, not at the border.

3.2 Technical architecture

Layer	Technology	Role
Consensus	SmartBFT (Fabric 3.0+)	Byzantine fault-tolerant ordering across organizations
Identity	Fabric CA + DSC (X.509)	Government officers sign with existing DSCs; no new PKI
Privacy	AES-256 envelope encryption; private data collections	Per-deal data isolation
Document storage	IPFS (encrypted, pinned)	Off-chain storage with on-chain hash anchoring
Middleware	Hyperledger FireFly	REST API layer for ICEGATE, GeM, and third-party integrations
Application	React + Node.js 20 LTS	Role-based web and mobile interfaces
Chaincode	Seven Smart Contracts	Deal lifecycle, quality, documents, payments, compliance, reputation, finance

3.3 Smart contract suite

Chaincode	Channel	Function
DealLifecycle	Trade	Full order state machine RFQ → PO → production → inspection → shipping → delivery.
DocumentManagement	Trade	Issues, transfers, and retires trade documents including e-BLs as unique digital tokens.
QualityVerification	Trade	Records inspector findings on-chain with DSC signature.
Compliance	Trade	Validates against DGFT policy, destination regulations, and sanctions. Rules updatable without chaincode upgrade.
PaymentTracking	Trade	Records payment milestones. No funds move on-chain.
Reputation	Network	Aggregates payment timeliness, quality acceptance, delivery reliability into access tiers.
SupplyChainFinance	Financing	OCEN-aligned loan lifecycle. Prevents duplicate financing..

3.4 Three-channel design

— **Network Channel.** Identity registry, reputation scores, compliance rules, and milestone hashes. All participants including banks and regulators.

— **Trade Channel.** Full deal lifecycle. Only hashes and encrypted payloads on-chain; no plaintext commercial data.

— **Financing Channel.** Banks, NBFCs, and regulators. Loan lifecycle isolated for regulatory separation.

3.5 Deep tech: oracles and analytics

- Factory IoT sensors (when available) can write production milestones directly to the chain.
- Current design extended with custom channels to accommodate sensor data.
- Transaction history across thousands of orders becomes the training dataset for anomaly detection models.

3.6 Security, scalability, performance

Metric	Value
Fabric SmartBFT throughput	~3,000 TPS
Peak demand at 1,000 concurrent orders	5–10 TPS
Headroom	300–600x
Document and QR verification latency	Sub-second
IPFS document retrieval	Under 5 seconds

KEY TAKEAWAY **Every MerkNet component is production-grade open source. The innovation is in the configuration, chaincode suite, and operational integration - not in inventing new infrastructure.**

4. Why Blockchain, Not a Database

Blockchain is the right primitive here not because it is fashionable, but because cross-border trade involves six to eight parties across jurisdictions who have no basis for trusting a single database operator.

Requirement	Centralized database	MerkNet
Multi-party trust without a central operator	No. Single point of trust failure.	Yes. Each party independently validates every record.
Signature verification without contacting the issuer	No. Manual callback required.	Yes. Cryptographic verification by any peer, instantly.
Immutability of issued documents	No. Admin can modify records.	Yes. Fabric blocks are cryptographically chained and timestamped.
Prevention of duplicate-BoL financing	No. No shared ownership record.	Yes. On-chain document registry enforces single ownership.
Non-repudiation of commitments	No. Disputes resolve on verbal evidence.	Yes. Every commitment is cryptographically signed.
Real-time regulator audit	Weeks to reconstruct history.	Regulators query their own peer directly.
Commercial confidentiality	Visible to the operator.	Envelope encryption: hashes visible, data hidden from non-parties.

KEY TAKEAWAY **The question is not "can a database do this?" It is "which party do we trust to operate it?" Blockchain eliminates the question.**

5. Prototype Approach

The Prototype will be built over four months and validated on five to ten real cross-border orders already in the Merkantis pipeline. It is structured as four technical milestones rather than a calendar schedule.

Milestone	Deliverable	Success criterion
M1 Network foundation	4-node Fabric 3.0+ network with FireFly middleware and IPFS cluster mimicking different roles and parties	Healthy consensus, sub-second commit, IPFS read/write verified
M2 Chaincode deployment	DealLifecycle and DocumentManagement chaincodes live with full test coverage	End-to-end flow from RFQ to document anchoring passes integration tests
M3 DSC integration	Class 3 DSC signing via Fabric CA; simulated DGFT and Customs identities	Signed Certificate of Origin verifiable by any peer in under 5 seconds
M4 Live orders	5–10 real India–Europe orders processed end-to-end; role-based UI; QR verification	All Section 6 metrics met on real orders

Why this approach is credible. Merkantis has an active supplier network in Coimbatore and Bangalore and active buyer relationships in Switzerland and Germany. The Prototype processes orders that would happen anyway - on-chain instead of by email. Government nodes are not required to demonstrate value; DGFT and Customs roles are simulated with separate MSP identities using valid / mock DSCs.

6. Government Alignment and Proliferation

MerkNet is not a blockchain searching for a government use case. It is the direct technical implementation of policies the Government of India has already enacted.

Policy or initiative	How MerkNet implements it
Foreign Trade Policy 2025	Mandates blockchain for trade records. MerkNet is the production implementation for manufacturing exports.
Bills of Lading Act 2025	Grants legal recognition to e-BLs. MerkNet's DocumentManagement chaincode is MLETR-aligned.
IT Act 2000 (S.4, S.5, S.65B)	DSCs are MerkNet's root of identity. Fabric transactions signed with DSCs carry legal weight.
MeitY National Blockchain Technology Stack	MerkNet is a candidate production workload for the national stack.
DPDP Act 2023	Data minimization by design: only hashes on-chain; PII encrypted off-chain; key destruction as erasure.
Atmanirbhar Bharat / Make in India	Reduces friction that hands Indian exporters a structural disadvantage.
District Export Hubs	Architecture replicates across all 700+ DEHs without modification.

6.1 Collaboration plan

Department	Ask	Stage
DGFT	Letter of intent for pilot; DSC signing test with officer credentials	Prototype
CBIC / Customs	ICEGATE API access for customs form pre-population	Prototype
MeitY	Hosting on National Blockchain Technology Stack	MVP
Ministry of MSME	Alignment with TIES scheme for digital trade infrastructure	MVP
RBI (Trade Finance)	Sandbox engagement for on-chain PO/invoice verification	MVP / Deployment
State industry depts (TN, KA)	CODISSIA and Karnataka MSME dept introductions	Prototype

Outreach in progress. Formal introductions via CODISSIA (Coimbatore) and CII networks. Target: letter of intent from at least one government department by end of Prototype stage.

6.2 Proliferation strategy

- **State rollout.** Tamil Nadu → Karnataka → Maharashtra → Gujarat, via state industry partnerships and DEH alignment.
- **Platform integration.** ICEGATE, GeM, Udyam via FireFly REST APIs. No platform migration for departments.
- **Association partnerships.** CII, FICCI, CODISSIA for MSME onboarding at scale.
- **Corridor expansion.** India–EU, then India–UAE, India–US, India–Africa.
- **API ecosystem.** Open APIs for third-party inspection, logistics, and finance applications.

KEY TAKEAWAY **Every proliferation step is a partnership activation, not a rebuild. The architecture is designed for the next 700 deployments, not just the first.**

7. Business Model

Revenue stream	Mechanism	Year 1 estimate
Transaction fee	Rs 2,000–10,000 per order anchored and verified	Rs 50–100L (1,000 orders)
Subscription	Monthly access for regular exporters and importers	Rs 5,000–20,000/month per participant
Managed infrastructure	Cloud node hosting for participants on self-managed identity	Rs 5,000–15,000/month per node
Finance facilitation	Per-verification fees from banks for on-chain PO/invoice checks	Rs 500–2,000 per verification
Government licensing	Platform-as-a-service for state trade facilitation bodies	Per-state contract

Market. India's merchandise exports were $437 billion in FY2024. MSME cross-border documentation services represent an estimated $2-3 billion annual services market. Merkantis's addressable segment is the precision manufacturing export corridor: India to Europe, North America, and the Middle East.

Use of grant funding. Blockchain partner vendor development and licensing costs; cloud hosting and compute operations; legal counsel for controller-processor agreements; government outreach and MoU drafting

8. Compliance Declarations

8.1 Non-crypto blockchain

> *"MerkNet uses zero cryptocurrency, zero mining, zero tokens, and zero proof-of-work. This is a permissioned enterprise blockchain with no connection to cryptocurrency of any kind."*

Built on Hyperledger Fabric 3.0+ (Apache 2.0), Hyperledger FireFly 1.3+ (Apache 2.0), and IPFS/Kubo 0.27+ (MIT, for encrypted document storage only, with no Filecoin or token incentive layer).

8.2 Open-source bill of materials

Component	Version	License	Role
Hyperledger Fabric	3.0+	Apache 2.0	Core ledger, chaincode runtime, identity
Hyperledger FireFly	1.3+	Apache 2.0	REST middleware and document management
IPFS (Kubo)	0.27+	MIT	Encrypted off-chain document storage
Go	1.21+	BSD 3-Clause	Chaincode language
Node.js	20 LTS	MIT	Application and API layer

Merkantis's original contributions - the seven-package chaincode suite, the progressive identity model, the envelope encryption layer, UCAN-based authorization, and DSC integration - are the core innovation. MerkNet is not a repackaged open-source project.

8.3 Regulatory compliance

Regulation	Compliance approach
IT Act 2000 (S.4, S.5, S.65B)	DSC-signed Fabric transactions carry legal weight as electronic records.
Bills of Lading Act 2025	e-BL implementation aligned with MLETR for cross-jurisdiction interoperability.
Foreign Trade Policy 2025	Direct implementation of the blockchain-for-trade-records mandate.
DPDP Act 2023	Data minimization by design; PII encrypted off-chain; key destruction as erasure.
Customs Act / ICEGATE	FireFly REST integration; pre-submission validation via Compliance chaincode.
RBI Digital Lending Directions 2025	Financing channel designed to comply; formal LSP registration is after initial deployment.
GDPR	Encrypted off-chain storage; controller-processor agreements between Indian and European entities.
UNCITRAL MLETR	e-BL implementation follows MLETR for adopting jurisdictions.
ICC Digital Standards Initiative	Document schemas aligned with ICC DSI standards for international trade.
WCO Data Model	Customs data aligned with WCO standards for cross-border interoperability.

8.4 IPR and originality

Merkantis acknowledges that IP developed under this challenge shall be held jointly by MeitY and Merkantis. MeitY's right to exercise IPR if commercialization is not initiated within 12 months is understood and accepted; Merkantis commits to active commercial deployment by Month 18. MerkNet is an original work; no copyright violation exists. The originality declaration on hundred-rupee Non-Judicial Stamp Paper will be provided per T&C requirements.

9. Why Merkantis Will Deliver

> *"MerkNet is built by a team that already executes cross-border manufacturing orders daily. The blockchain layer digitizes workflows we already run manually."*

That is the single sentence that separates this submission from the rest of the field. Most blockchain pilots fail for the same reason: the team building the platform does not know the business it is digitizing. MerkNet reverses that failure mode. Merkantis is an operating export execution company; the blockchain is the tool, not the purpose.

- **Operational credibility.** Merkantis executes cross-border precision manufacturing procurement between Indian suppliers and European buyers today. We know where documents break because we have processed real orders through broken processes.
- **Zero cold-start.** The Prototype runs on orders already in our pipeline. Suppliers in Coimbatore and Bangalore; buyers in Switzerland and Germany. No user acquisition, no synthetic data.
- **Technical depth.** Founder with BITS Pilani engineering background and applied experience in B2B product, sales, and operations. Hyperledger Fabric development handled by partners at khaaliTech Softwares Pvt. Ltd., founded by an Ivy League graduate.
- **Government and industry access.** Director-level advisor with decades of multinational industrial leadership in India, active in the Swiss-Indian Chamber of Commerce (Western Region), with direct access to CODISSIA, CII, and state industry departments.

9.1 Where other platforms failed

Dimension	TradeLens / Marco Polo	IndiaChain / Govt Pilots	MerkNet
Scope	Shipping or finance only	Single-department PoCs	Full lifecycle: RFQ through payment
Status	Both defunct (2022–2023)	Narrow pilots, limited scale	Building on a live trade pipeline
Onboarding	Required own infrastructure	Department-specific onboarding	Zero-infrastructure; managed identity
National PKI	No DSC integration	Partial	DSC-native from day one
India focus	None	Yes but narrow	Built for Indian context end-to-end
Manufacturing quality	Not covered	Not covered	QC workflow and inspection anchoring built in

KEY TAKEAWAY **The team, the pipeline, and the stakeholders are already in place. A grant unlocks increased prioritisation on the infrastructure and integration.**

10. Single Point of Contact

Field	Detail
Name	Divyansh Malhotra
Designation	Founder, Merkantis Private Limited
Email	divyansh@merkantis.com
Mobile	9928001302
Entity	Merkantis Pvt Ltd
Registration	Indian Startup per DPIIT guidelines; not a subsidiary of any foreign corporation
DPIIT number	[to be added]

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Merkantis Private Limited